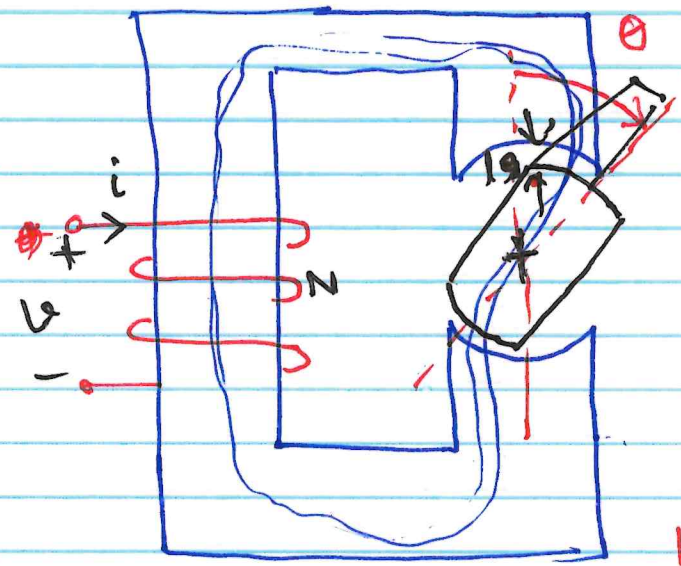
Lecture 18

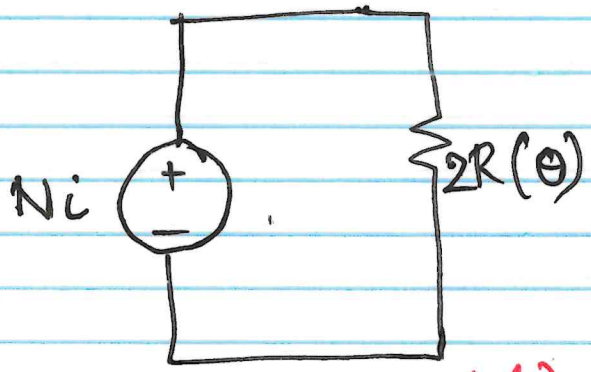


$L(\theta) =$

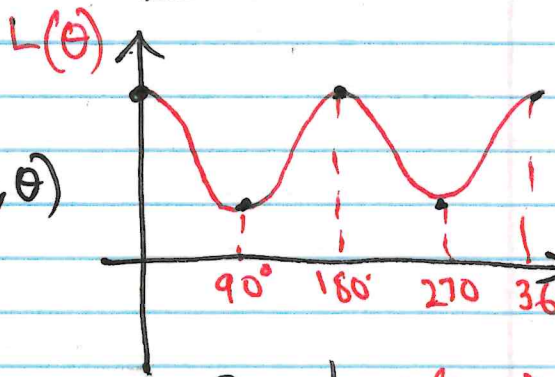
$\theta = 0°, 360°$, max 180

$L(\theta) \frac{N^2}{R(\theta)}$

$$R(\theta) = \frac{l_g(\theta)}{\mu_0 A_g(\theta)}$$



$$L(\theta) = L_1 + L_2 \cos 2\theta$$



$\lambda = L(\theta) \cdot i$

$\omega_m(\lambda, \theta)$

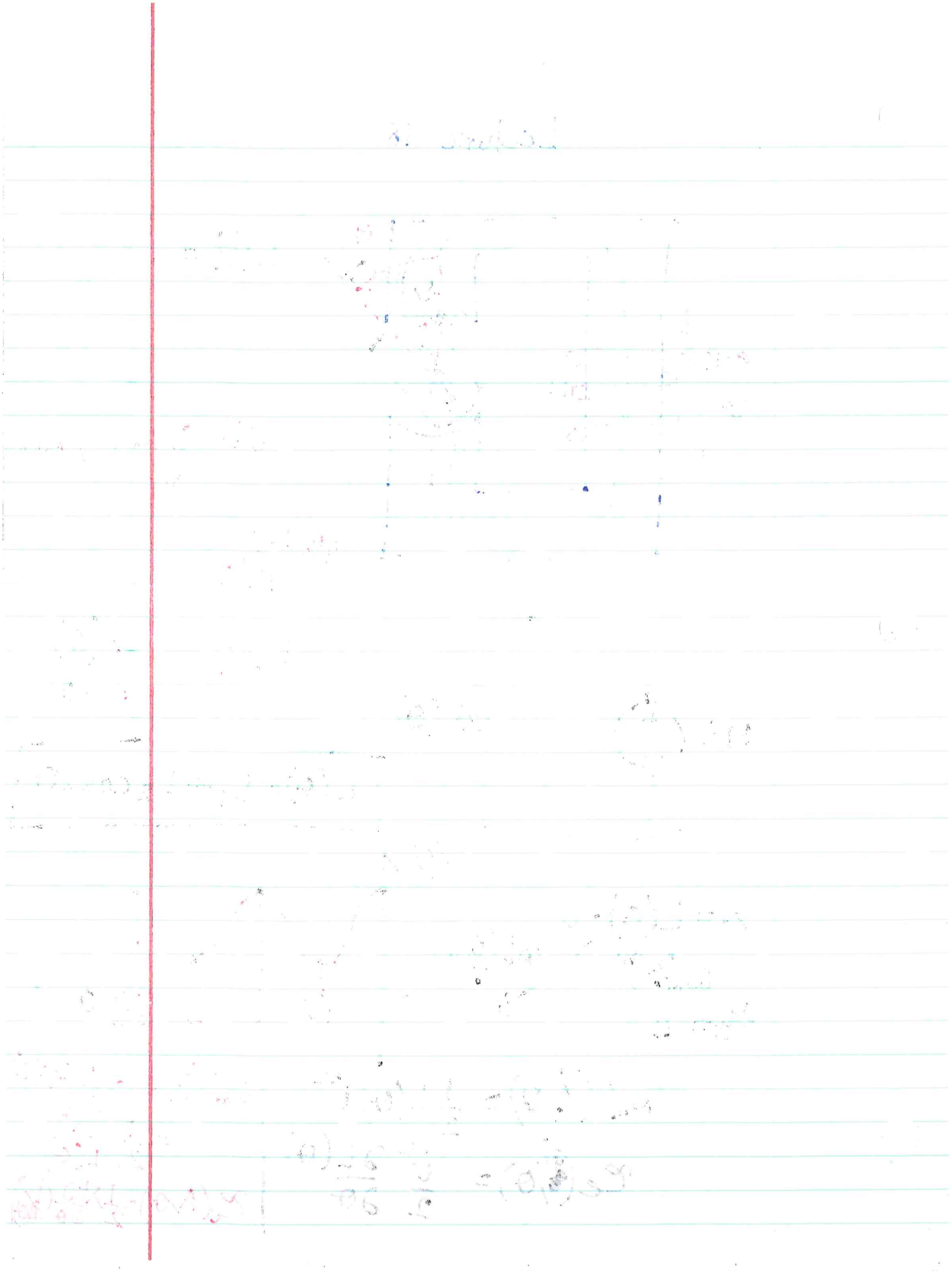
$\omega_m'(i, \theta)$

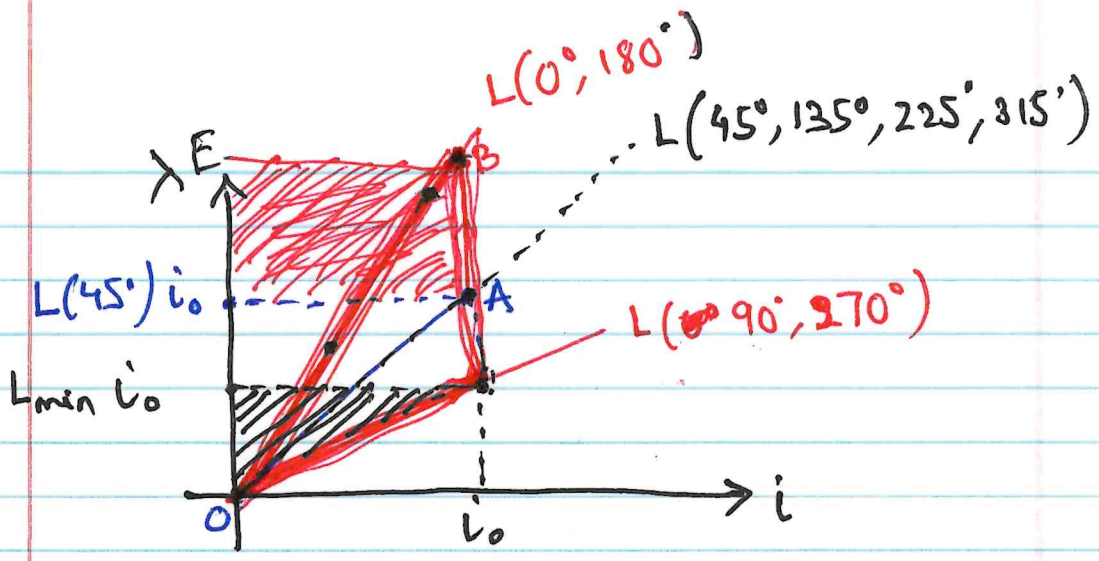
$$\omega_m'(i,\theta) = \frac{1}{2} L(\theta) i^2$$

$$\tau_e(i,\theta) = \frac{i^2}{2} \frac{\partial L(\theta)}{\partial \theta}$$

$$\omega_m(\lambda,\theta) = \frac{1}{2} L(\theta) i^2 = \frac{1}{2} \frac{\lambda^2}{L(\theta)}$$

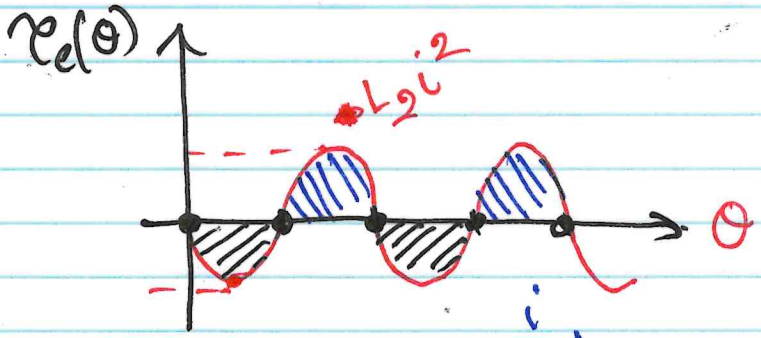
$$\tau_e(\lambda,\theta) = \frac{1}{2} \lambda^2 \frac{\partial}{\partial \theta}\left(\frac{1}{L(\theta)}\right)$$

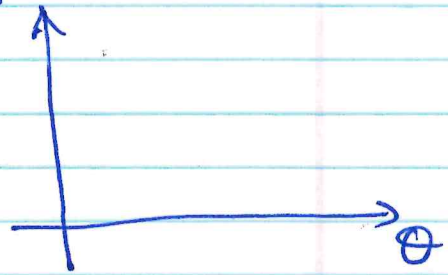




$$\tau_e(i,\theta) = -\frac{i^2}{2}\frac{dL_2}{d\theta}\sin 2\theta$$

$$= -\frac{L_2 i^2}{4}\sin 2\theta$$





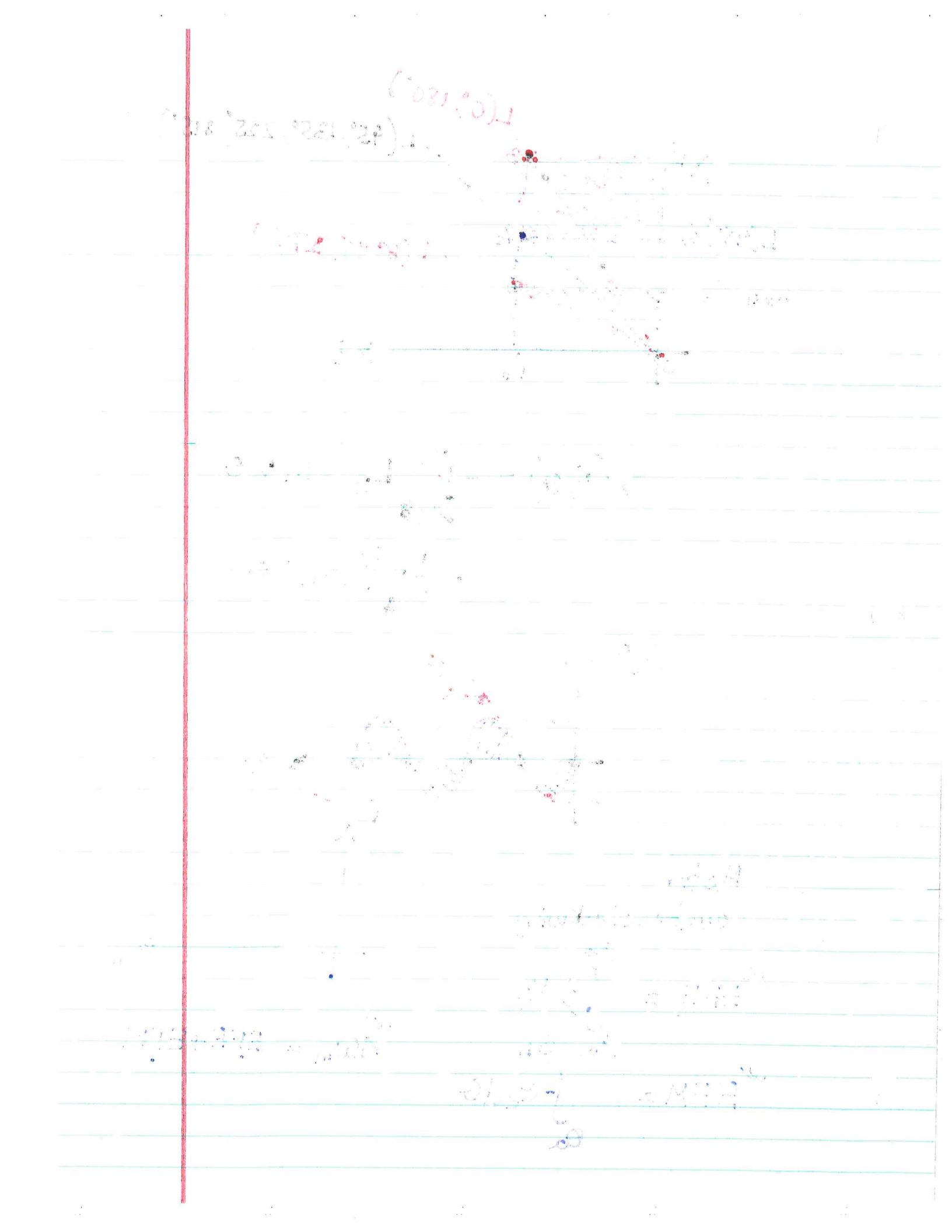
Motor.

counterclockwise

* $EFE = \int_{\lambda_a}^{\lambda_b} i\,d\lambda$

* $EFM = \int_{\theta_a}^{\theta_b} -\tau_e\,d\theta$

* $\Delta\omega_m = EFE + EFM$



$0 \rightarrow A$

$$W_m(A) = \int_0^{L(45^\circ) i_0} \frac{\lambda'}{L(45^\circ)} \cdot d\lambda' = \frac{1}{2} L(45^\circ) i_0^2$$

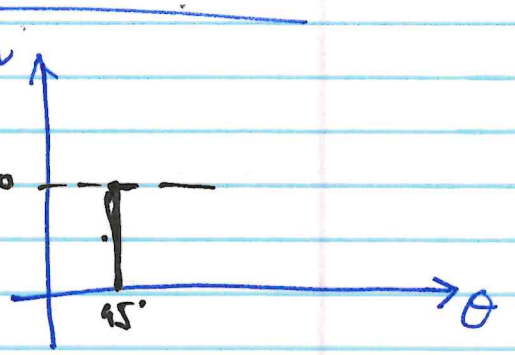
$$\underline{EFE} = \int_0^{i_0} i \, d\lambda = \frac{1}{2} L(45^\circ) i_0^2$$

$$\underline{EFM} = 0$$

$A \rightarrow B$

$$EFE = \int_{\lambda_a}^{\lambda_b} i \, d\lambda = i(\lambda_b - \lambda_a)$$

EFM



$$\Delta W_m = W_m(B) - W_m(A)$$

$$= \frac{1}{2} L(0^\circ) i_0^2 - \frac{1}{2} L(45^\circ) i_0^2$$

$$EFM = -\int_{\theta_a}^{\theta_b} -L_2 i_0^2 \sin 2\theta \, d\theta =$$

$$\oint_C \Delta\omega_m = \oint EFE + \oint EFM$$

$$0 = \oint_C EFE + \oint_C EFM$$

$\oint EFE > 0$ and $\oint EFM < 0$ } Motor

$\oint EFE < 0$ and $\oint EFM > 0$ } Generate